

Find the angle θ between the vectors $u = (4, -3)$ and $v = (1, -2)$

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}$$

$$1.) u \cdot v = 4(1) + (-3)(-2) = 4 + 6 = 10$$

$$2.) \|u\| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$$

$$3.) \|v\| = \sqrt{(1)^2 + (-2)^2} = \sqrt{1 + 4} = \sqrt{5}$$

$$4.) \cos \theta = \frac{10}{5\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$5.) \theta = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) \approx 0.464 \text{ radians}$$